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What is the influence of psychosocial factors on artificial intelligence appropriation in college students?

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Abstract

Background In recent years, the adoption of artificial intelligence (AI) has become increasingly relevant in various sectors, including higher education. This study investigates the psychosocial factors influencing AI adoption among Peruvian university students and uses an extended UTAUT2 model to examine various constructs that may impact AI acceptance and use.

Method This study employed a quantitative approach with a survey-based design. A total of 482 students from public and private universities in Peru participated in the research. The study utilized partial least squares structural equation modeling (PLS-SEM) to analyze the data and test the hypothesized relationships between the constructs.

Results The findings revealed that three out of the six hypothesized factors significantly influenced AI adoption among Peruvian university students. Performance expectancy ($\beta = 0.274$), social influence ($\beta = 0.355$), and AI learning self-efficacy ($\beta = 0.431$) were found to have significant positive effects on AI adoption. In contrast to expectations, ethical awareness, perceived playfulness, AI readiness and AI anxiety did not have significant impacts on AI appropriation in this context.

Conclusion This study highlights the importance of practical benefits, the social context, and self-confidence in the adoption of AI within Peruvian higher education. These findings contribute to the understanding of AI adoption in diverse educational settings and provide a framework for developing effective AI implementation strategies in higher education institutions. The results can guide universities and policymakers in creating targeted approaches to enhance AI adoption and integration in academic environments, focusing on demonstrating the practical value of AI, leveraging social networks, and building students' confidence in their ability to learn and use AI technologies.

Keywords Artificial intelligence, AI appropriation, Higher education, UTAUT2 model, Peruvian university students, Technology adoption

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Introduction

Artificial intelligence (AI) is significantly transforming higher education on a global scale, revolutionizing traditional educational processes and offering new learning opportunities. AI applications in this field range from task automation to personalized learning experiences, which impact teaching, learning, and organizational management [1, 2]. Key AI applications in higher education include adaptive learning systems, personalized learning experiences, and intelligent virtual environments [1, 3]. These advances are shifting traditional educational paradigms, moving the focus from conventional classrooms to AI-enhanced learning environments that can improve student outcomes and optimize administrative tasks [4, 5].

However, the integration of AI into higher education is not without challenges. Ethical concerns arise regarding data privacy, algorithmic bias, and the possibility that AI could replace educators [1, 6, 7]. Furthermore, issues surrounding global inequality and the unequal distribution of AI resources have also been raised [2, 4, 8].

Despite these challenges, AI's potential to democratize and personalize learning, address persistent educational problems, and reduce overall education costs is significant [8, 9]. Nevertheless, it is crucial to develop comprehensive ethical guidelines to ensure that AI implementation in higher education aligns with the fundamental values of academic integrity and social responsibility.

The adoption of AI in higher education is influenced by a complex interplay of psychosocial factors, as revealed by recent literature. Studies have identified perceptions of utility and effectiveness as key determinants [9, 10], along with ethical and privacy considerations, which may act as barriers to their implementation [10, 11]. The impact of AI on students' psychological well-being presents a duality, offering personalized support but also generating potential stress and anxiety [12]. Learning outcomes and cost effectiveness emerge as influential factors in student acceptance [11], whereas the adaptability and personalization of AI tools are considered pivotal drivers [13]. Cognitive trust in educational AI, shaped by transparency, reliability, and ethical considerations, plays a significant role in its adoption [14]. Contextual factors such as performance expectations, effort expectations, social influence, and facilitating conditions also impact the intention to use and user behavior [15, 16]. Students' attitudes and perceptions toward these technologies significantly affect their acceptance and use [15]. Importantly, these factors interact in complex ways, and their relevance may vary depending on the educational and cultural context [17], highlighting the need for more extensive research to fully understand the dynamics of AI adoption in diverse university settings.

Nevertheless, despite various studies on the psychosocial factors involved in university students' adoption of artificial intelligence (AI), significant knowledge gaps remain. A study with Peruvian university professors revealed a considerable gap in knowledge about AI and its educational application [18], suggesting that this lack of knowledge may extend to students, potentially limiting their ability to use AI tools effectively. While research in other contexts has identified factors such as epistemic capability, enabling environments, and psychological attitudes as critical for AI adoption [19], a comprehensive understanding of the interaction and variability of these factors in different cultural and educational contexts is lacking. Moreover, although challenges such as a lack of technical knowledge, privacy concerns, and unequal access to AI resources have been recognized, further exploration is needed to understand their influence on long-term adoption. The educational implications and impact on students' psychological well-being, which range from personalized support to the generation of stress [12], represent another area where current knowledge is limited. Thus, this gap underscores the need for more exhaustive and longitudinal studies addressing the complexity of psychosocial factors in the adoption of AI by university students in diverse global contexts.

To address these knowledge gaps, this study employs the UTAUT2 model as its theoretical foundation, extending it to incorporate psychological and social factors specific to AI adoption in higher education [15, 16]. The UTAUT2 model was selected because it integrates hedonic motivation and social influence, making it particularly suitable for understanding the complex psychological dynamics of AI adoption among students [10, 11]. The model's flexibility enables the integration of context-specific constructs, such as SL and EA, which are crucial in educational settings [12, 13]. The comprehensive framework of UTAUT2 facilitates the examination of both individual psychological factors and social influences, providing a more nuanced understanding of AI adoption behavior [14, 15]. Research has demonstrated the model's strong explanatory power across different cultural contexts and technologies [16, 17], making it well suited for studying AI adoption in the Peruvian higher education context. The extension of UTAUT2 in this study to include psychological constructs such as AI anxiety and EA addresses the gaps in understanding how emotional and ethical considerations influence AI adoption [10, 12], whereas the inclusion of SI factors acknowledges the collective nature of technology adoption in educational settings [15, 16]. This adapted theoretical framework enables a comprehensive examination of how psychological readiness, social dynamics, and ethical considerations interact to influence AI adoption among university students [17, 18].

In this study, the following constructs are considered: performance expectancy (PE), social influence (SI), perceived playfulness (PP), ethical awareness (EA), AI learning self-efficacy (SL), AI readiness and AI anxiety (AAN), and AI appropriation (AIO).

The aim of the present study is to analyze the factors influencing the adoption of AI by university students in the Peruvian context. The general research question guiding this study is as follows: What psychosocial factors influence the appropriation of artificial intelligence among Peruvian university students? The specific research questions are as follows: (1) How does PE influence AIO among college students? (2) How does SI influence AIO in college students? (3) How does PP influence AIO in university students? (4) How does EA influence AIO among university students? (5) How does self-efficacy in learning AI influence AIO among college students? (6) How does AAN influence AIO in college students? From a theoretical perspective, this study is justified by the existence of significant knowledge gaps in understanding how university students adopt AI in diverse educational contexts. From a theoretical perspective, this study is justified by the existence of significant knowledge gaps in understanding how university students adopt AI in diverse educational contexts. While previous research has identified factors such as epistemic capability, enabling environments, and psychological attitudes as critical for AI adoption, a comprehensive understanding of the interaction and variability of these factors in different cultural and educational contexts, particularly in the Peruvian context, is lacking. This research contributes to filling that theoretical gap by providing a completer and more contextualized conceptual framework.

In practical terms, this study is justified by its potential to inform and improve AI implementation strategies in Peruvian higher education. Identifying the psychosocial factors that influence students' adoption of AI would allow educational institutions to develop more effective approaches to integrate these technologies, addressing challenges such as the lack of technical knowledge, privacy concerns, and unequal access to AI resources. Additionally, the findings could guide the design of more effective training and support programs for students and faculty. From a social perspective, this research is justified by its potential to contribute to the democratization and personalization of learning in Peruvian higher education. Understanding the psychosocial factors involved in AI adoption could help reduce inequalities in access to and use of these technologies, promoting a more equitable education that meets individual student needs. Moreover, by addressing the educational implications and impact on students' psychological well-being, this study could contribute to a more ethical and responsible use of AI in higher education, aligned with the values of academic integrity and social responsibility.

This study makes several significant contributions to the field of AI adoption in higher education. First, it provides a theoretical contribution by extending the UTAUT2 model through the incorporation of new variables specifically tailored to AI adoption in Peruvian higher education, such as SL and EA. This extension enhances our understanding of technology acceptance models in emerging educational contexts. Second, the study offers a methodological contribution by developing and validating a comprehensive instrument to measure AIO among university students, which can be adapted and used in similar educational settings. Third, it makes a practical contribution by providing empirical evidence that can guide policymakers and educational administrators in developing effective strategies for AI implementation in higher education institutions, particularly in the context of developing countries. Finally, this research contributes to the growing body of knowledge about psychosocial factors influencing AI adoption by offering insights from a Latin American perspective, thus adding a unique cultural dimension to the literature predominantly focused on Western and Asian contexts. This multicultural perspective enriches our understanding of how cultural and contextual factors may moderate the adoption of advanced technologies in educational settings.

Literature review

Review of the latest perspectives on AI immersion in higher education

Personalized learning has emerged as one of the most promising applications of AI in higher education. Research indicates that AI facilitates the creation of customized learning pathways, significantly enhancing student engagement and academic outcomes [20, 21] [22]. supports this view, noting that adapting content to individual student needs represents a paradigm shift in pedagogy. However [23], cautions against overreliance on technology, emphasizing that human interaction remains essential to the educational process.

In the realm of administration, AI involves streamlining processes within higher education institutions [24]. reported that automating administrative tasks through AI improves operational efficiency, enabling educators to devote more time to teaching and research [25]. echoes this sentiment, highlighting the transformative potential of AI for educational management. However [9], warns that unequal access to these technologies could exacerbate existing disparities between resource-rich and resource-poor institutions.

AI has also had a notable effect on global access and equity in higher education [19]. demonstrated how AI helps to overcome geographical barriers, democratizing

access to high-quality education [26]. further emphasized AI's potential to foster educational inclusion. Nevertheless [27], highlights the risk of new forms of digital exclusion if disparities in access to technology are not adequately addressed.

AI-driven teaching innovations are reshaping classroom dynamics [28]. illustrates how adaptive learning systems and smart classrooms create more interactive learning environments [29]. supports this view, stressing AI's capacity to develop critical 21st-century skills. However [30], argues that AI integration must be balanced with traditional methods to preserve the valuable aspects of human interaction in education.

The implementation of AI in higher education presents significant challenges, particularly with respect to ethics and academic integrity [28]. underscores the need for accuracy, fairness, and transparency in AI algorithms. Similarly [31], stressed the importance of fostering AI literacy among students and educators. On the other hand [32], argues that the benefits of AI in detecting plagiarism and promoting academic integrity outweigh the potential risks.

To navigate these challenges and maximize benefits, scholars recommend a balanced approach [20]. advocated updating curricula and providing AI literacy training [33]. also emphasized the importance of preparing educators and students for an AI-driven future. However [27], advised caution, calling for more research on the long-term effects of AI in higher education.

In conclusion, AI's integration into higher education presents a unique combination of opportunities and challenges. While researchers such as [20] and [24] underscore the transformative potential of AI, others, including [9] and [27], highlight the associated risks. The prevailing consensus suggests that successful AI integration will depend on a balanced strategy that maximizes benefits while mitigating risks, ensuring that AI serves to enhance the quality and accessibility of higher education globally.

Psychosocial factors influencing AI appropriation in college students

PE refers to the degree to which individuals believe that using AI technology will enhance their academic performance and productivity [34]. In educational contexts, PE has been identified as a fundamental driver of technology acceptance, as students evaluate the potential benefits of AI tools in improving their learning outcomes and academic efficiency [35].

SI represents the extent to which students perceive that important others, including peers, professors, and educational institutions, believe that they should use AI technologies [35, 36]. The social environment within educational institutions plays a crucial role in shaping students' attitudes and behaviors toward AI adoption [10].

PP, also known as hedonic motivation, reflects the fun, enjoyment, or pleasure derived from using AI technology [36]. In educational settings, the playful aspects of technology interaction can significantly influence students' willingness to engage with and adopt AI tools for learning purposes [15].

EA encompasses students' understanding and consciousness of the ethical implications and responsibilities associated with AI use in academic contexts [10]. This construct has gained importance as educational institutions grapple with questions of academic integrity and responsible AI use [15, 16].

SL refers to students' confidence in their ability to learn and effectively use AI technologies [37]. This construct is particularly relevant in educational settings, as it influences students' persistence and effort in mastering new AI tools [34, 35].

AANs represent two interrelated aspects: preparedness to adopt AI technologies and concerns or apprehensions about their use [37]. Readiness reflects students' psychological and technical preparedness to integrate AI into their learning practices, whereas anxiety captures the emotional and psychological barriers that might hinder adoption [10, 15].

AIO represents the extent to which students incorporate and integrate AI technologies into their educational practices [38]. This construct goes beyond mere acceptance to encompass how students actively adapt and utilize AI tools to support their learning objectives [34, 38].

These constructs interact dynamically within the educational environment, influencing how students adopt and integrate AI technologies into their academic practices [15, 16]. Understanding these interactions is crucial for developing effective strategies to promote successful AI integration in higher education [37, 38].

Scientific support of the research hypotheses

PE has been shown to significantly influence AIO among university students, as evidenced by several recent studies. Research has revealed that PE positively affects students' willingness to accept AI-assisted learning environments [34], which is a clear indicator of AIO. Additionally, PE significantly influences students' behavioral intentions to use generative AI products [39], another crucial aspect of AIO. This relationship is reinforced by findings that PE is an important predictor of the acceptance and use of AI tools in higher education [40], directly aligning with the concept of AIO.

The strength of the relationship between PE and AIO has been confirmed across various contexts, including a study on the use of ChatGPT among university students, which demonstrated that PE significantly influences students' behavioral intentions to use this AI tool [16]. From a broader perspective, research has extended the

UTAUT2 model to include ethical factors, finding that PE has a positive influence on students' behavioral intentions to use generative AI products [39]. These consistent findings across multiple studies and contexts [16, 34, 39, 40] provide robust support for the hypothesis that PE significantly influences AIO among university students, although it is important to note that the studies covered different AI applications and contexts. Therefore, we propose the following:

Hypothesis 1 PE significantly influences AIO in college students.

A study involving South Indian university students revealed that SI has a significant direct positive effect on the intention to use AI-enabled job application processes [41]. This finding suggests that the social environment of college students can strongly influence their willingness to adopt and appropriate AI technologies, which is a key aspect of AIO. The impact of SI on AIO is further supported by research demonstrating how students adapt their behaviors in response to AI interactions. A mixed-methods study with young individuals revealed that participants adjusted their behaviors to complement different types of AI teammates [42]. This behavioral adaptation highlights the powerful role of SI in shaping how students interact with and appropriate AI technologies.

Moreover, the influence of social factors on AIO is evident in the context of AI-based education. Research in the U.S. highlighted the crucial role of perceived social presence in AI instructors, showing that students with high expectations for instructor nonverbal immediate behaviors demonstrate more positive perceptions of AI-based education when they experience stronger social presence in their AI instructor [43]. This finding underscores the importance of social cues and expectations in students' acceptance and appropriation of AI in educational contexts.

The relationship between SI and AIO is also reflected in broader STEM integration. Longitudinal research with students from historically overrepresented groups in STEM has revealed that interactions with SI agents, such as faculty mentor support and research engagement, promote integration into the STEM community through the development of science identity and community values [44]. This suggests that SI can play a significant role in shaping students' engagement with and appropriation of advanced technologies, including AI.

Additionally, research on the factors influencing academicians' intentions to continue using AI-based chatbots at Indian universities has highlighted the nuanced role of peer networks in shaping adoption [15]. This finding further supports the idea that SI significantly affects how college students appropriate and integrate AI technologies into their academic lives. While these studies [15, 30, 41, 42, 44] provide strong support for the hypothesis that SI significantly influences AIO among college students, they cover various applications and contexts of AI. However, the consistency of findings across multiple studies and contexts offers robust support for the proposed hypothesis.

Hypothesis 2 SI significantly influences AIO in college students.

PP (PP) has emerged as a significant factor in AIO among university students. The integration of PP into the UTAUT model for AI-based learning platforms underscores its importance in understanding students' behavioral intentions toward AI technologies [45]. This inclusion suggests that the playful aspects of interactions with AI can significantly influence how students appropriate and engage with AI tools in their learning environments.

Research has shown that incorporating game elements to increase PP positively impacts student participation. Specifically, the use of badges has demonstrated a positive relationship with PP [46], indicating that enhancing the playful aspects of AI technologies could lead to greater appropriation and engagement among university students.

Moreover, gender differences have been observed in the effects of playfulness on students' attitudes toward technology use, with playfulness directly influencing female students' attitudes [47]. This finding highlights the nuanced role of PP in AI appropriation, suggesting that its impact may vary across different demographic groups within the student population.

The importance of PP in AIO is further reinforced by research on AI-based education. A study highlighted the crucial role of perceived social presence in AI instructors, indicating that students with high expectations of nonverbal immediacy behaviors from their instructors have more positive perceptions of AI-based education [43]. While this study does not directly measure PP, it suggests that the interactive and engaging aspects of AI technologies, which are closely related to PPs, can significantly influence students' perceptions and, by extension, their appropriation of AI in educational contexts. On the basis of this evidence, although further direct research on the specific relationship between PP and AIO in university students is needed, the following hypothesis is proposed:

Hypothesis 3 PP significantly influences AIO in university students.

EA has been shown to significantly influence AIO among university students, as evidenced by several recent studies. Research has revealed that university students' EA significantly affects their behavioral intentions and actual use of generative AI products [39]. This finding suggests

that understanding the ethical implications of AI plays a crucial role in how students adopt and appropriate these technologies.

Furthermore, EA can positively influence students' intentions to use generative AI products, although it may also heighten their perceptions of ethical risk [39]. This duality underscores the complexity of the relationship between EA and AI appropriation, indicating that greater awareness may both encourage and moderate AI use, depending on individual ethical evaluations.

The importance of EA in AIO is reinforced by studies on AI literacy programs. One study showed that an AI literacy program successfully improved participants' EA, emphasizing the importance of ethical considerations in AI education [48]. This finding suggests that AI ethics education can directly influence how students perceive and appropriate AI technologies. Exposure to ethical instruction and internship experiences has also been shown to influence communication students' ethical perceptions, including awareness of AI-related ethical issues [49]. Moreover, an explicit-reflective online learning module significantly improved graduate science and engineering students' knowledge of AI ethics and their ability to identify and articulate ethical issues in AI [50]. These results indicate that ethical education can have a direct effect on students' understanding and, consequently, their appropriation of AI.

Concerns about privacy, ethics, social factors, and academic resources significantly influence AI adoption among university students [10], underscoring the importance of EA in the AIO process. Furthermore, the ethical challenges associated with AI and the Internet of Things pose risks to privacy and data security, highlighting the need for education on ethical issues related to technology [51]. On the basis of this evidence, although further direct research on the specific relationship between EA and AIO in university students is needed, the following hypothesis is proposed:

Hypothesis 4 EA significantly influences AIO among university students.

SL has been shown to significantly influence AIO among university students, as evidenced by several recent studies. Research indicates that self-efficacy in AI can positively influence students' attitudes toward AI and their actual use of AI tools [37, 52]. This finding suggests that students' confidence in their ability to learn and use AI plays a crucial role in how they adopt and appropriate these technologies.

Additionally, AI self-efficacy has been observed to have an indirect positive effect on the learning effectiveness of AI-based technological applications [53]. This finding indicates that students with higher self-efficacy tend to make better use of AI tools in their learning processes, likely leading to greater appropriation of these technologies.

The importance of self-efficacy in AIO is reinforced by studies demonstrating that self-efficacy in learning positively affects learning intentions in the AI context [52, 54]. This relationship suggests that students with greater confidence in their AI learning abilities are more likely to engage with these technologies, leading to greater appropriation.

Factors such as attitudes toward AI learning, confidence in AI learning, and subjective norms have been shown to significantly influence students' intentions to learn AI [55]. These findings underscore the complexity of the relationship between self-efficacy and AI appropriation, indicating that multiple interrelated factors contribute to the appropriation process.

In the context of translation technologies, a study demonstrated a positive prediction of perceived ease of use and enjoyment from computer self-efficacy, which increased students' attitudes and behavioral intentions to use translation technologies [38]. While this study focuses on translation technologies, it provides relevant insights into how self-efficacy may influence the adoption and appropriation of AI-based technologies in general.

Importantly, the influence of self-efficacy may vary by gender and learning environment. One study revealed that the impact of campus learning environments on self-directed learning self-efficacy was more significant for men than for women [56], suggesting the need to consider contextual factors when examining the relationship between SL and AIO. On the basis of this evidence, although further direct research on the specific relationship between SL and AIO in university students is needed, the following hypothesis is proposed:

Hypothesis 5 SL significantly influences AIO in college students.

AANs have been shown to significantly influence AIO among university students, as evidenced by several recent studies. Research indicates that AI readiness, confidence, and the perceived relevance of AI positively influence students' willingness to learn about AI [57]. This finding suggests that the level of readiness and perception of AI importance play key roles in how students approach and potentially appropriate these technologies.

However, the relationship between AI anxiety and appropriation is more complex. One study revealed that AI learning anxiety negatively affects learning motivation, whereas job replacement anxiety related to AI has a positive effect on extrinsic motivation [52]. This finding indicates that different types of AI-related anxiety can influence students' willingness to engage with AI in distinct ways.

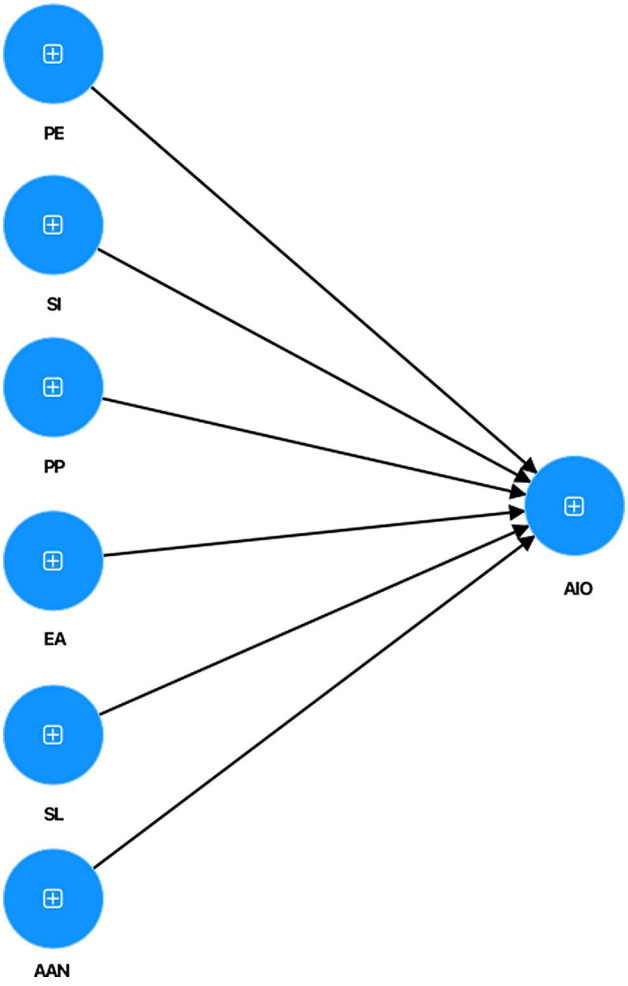

Fig. 1 Proposed research model. Note: Performance expectancy = PE; SI = Social influence; PP = Perceived playfulness; EA = Ethical awareness; AI learning self-efficacy = SL; AAN = AI readiness and AI anxiety; AIO = AI appropriation

The importance of AI readiness and anxiety in AIO is further reinforced by a study on medical students, which revealed an inverse relationship between AI readiness and anxiety [58]. This finding underscores the importance of increasing students' preparedness for AI applications and reducing their anxieties as a means of facilitating AI appropriation.

Additionally, AI anxiety has been reported to negatively predict the actual use of AI tools among L2 university students [37]. This result indicates the practical impact of anxiety on engagement with AI resources, suggesting that lower anxiety levels could lead to greater AI appropriation (Fig. 1).

Importantly, gender differences exist in AI readiness and anxiety. Compared with female students, male students reported greater confidence, perceived relevance, and readiness for AI [57]. This suggests the need to consider demographic factors when examining the relationship between AAN and AIO. On the basis of this evidence, although further direct research on the specific relationships among AI readiness, AI anxiety, and AIO in university students is needed, the following hypothesis is proposed:

Table 1 Sociodemographic characteristics of the sample

Category	Item	*N*	%
Gender	Male	281	58.3
	Female	201	41.7
Age	24–28	153	31.8
	29–33	183	38.07
	34–38	132	27.48
	39 to more	13	2.64
Type of university	Public	300	62.33
	Private	182	37.7
Level of studies	Graduate student	286	59.3
	Teacher/senior researcher	27	5.7
	Postdoctoral researcher	6	1.3
	Research Administrator	1	0.3
	Independent researcher	161	33.3
AIOtime	1–2 Months	170	35.2
	3–5 Months	120	24.9
	5–12 Months	86	17.9
	1–2 years	64	13.3
	More than 3 years	41	8.6

Hypothesis 6 AANs significantly influence AIO in college students.

Methods and materials

To test the research hypotheses, the researchers conducted an empirical assessment on the basis of the work of [59]. A survey was administered to university students with experience using AI models to gather data on their perceptions and attitudes toward these technologies.

Participants

The study involved 482 university students from public and private institutions in Peru. The sample was selected through nonprobabilistic convenience sampling, as described by [60]. This method allowed for the inclusion of participants who were available and willing to voluntarily contribute to the research. Although this type of sampling does not guarantee full representativeness of the university student population, it was appropriate for this exploratory study, whose primary goal was to identify preliminary trends and patterns rather than to make broad generalizations.

According to Tables 1 and 58.3% of the participants were male (281), whereas 41.7% were female (201). In terms of age distribution, the largest group was in the 29–33 years age range, accounting for 38.07% (183 participants), followed by the 24–28 years age range,

accounting for 31.8% (153 participants). The majority of the respondents were from public universities (62.33%, 300 participants), whereas 37.7% (182 participants) were from private institutions.

In terms of educational level, postgraduate students predominated, representing 59.3% (286 participants) of the sample. They were followed by independent researchers, who made up 33.3% (161 participants) of the sample. With respect to the duration of AI use, most participants (35.2%, 170 participants) reported having used AI for 1–2 months, followed by those with 3–5 months of experience (24.9%, 120 participants).

Instruments

A self-developed instrument was used to collect the data for the study (see supplementary file). This instrument is based on the literature and the constructs of the UTAUT2 model, which also incorporates concepts of perceived ethics and academic integrity. The questionnaire was developed through Google Forms and was structured in three main sections.

The first section contained informed consent, provided detailed information about the study and ensured the anonymity of the participants. A branching question was included to confirm voluntary participation. The second section collected sociodemographic data, including age, gender, type of university, education level, and duration of AI usage.

The third section included evaluation items designed to measure the constructs of the proposed model. This section included a total of 37 items distributed across the following constructs: PE, SI, PP, EA, SL, AAN, and AIO. All the items were measured via a 5-point Likert scale, where 1 represented "strongly disagree" and 5 "strongly agree."

Procedure and data analysis

The survey was conducted over a six-month period, from October 2023 to March 2024. The researchers obtained permission from universities and higher education institutions (HEIs) to distribute the survey online. The form was shared via email and distributed through student WhatsApp groups. For the analysis of the collected data, the researchers followed a systematic process that included several steps:

Initially, data cleaning was performed via Microsoft Excel, and missing values and incomplete surveys were removed to ensure the quality of the information. Subsequently, descriptive statistical techniques were applied to create the sociodemographic results table, providing an overview of the sample's characteristics. A confirmatory factor analysis (CFA) was conducted to evaluate the main reliability and validity indicators of the measurement model. In this process, indicators such as factor loadings and average variance extracted (AVE) were used. Additionally, to assess internal consistency reliability, Cronbach's alpha and composite reliability (CR) (rho_a and rho_c) were employed. Discriminant validity was evaluated via the [61] and the heterotrait–monotrait ratio (HTMT) criterion, ensuring adequate distinction between the model's constructs.

Finally, partial least squares structural equation modeling (PLS–SEM) was used to test the research hypotheses. This analysis was carried out via SmartPLS software.

Results

Results of the measurement model

Partial least squares structural equation modeling (PLS-SEM) was used in this study. Consequently, a confirmatory factor analysis (CFA) was carried out to confirm the convergent validity of the measurement model. Table 2 presents the factor loadings of the items, and according to [62], acceptable factor loadings higher than 0.70 are met; moreover, all the measured constructs present average variance extracted (AVE) values that exceed the threshold of 0.50, as proposed by [63]. On the other hand, the standard deviation of the items is between 0.008 and 0.092, which indicates that it seems that the items are not excessively dispersed with respect to their means.

Table 3 shows the results of the reliability and discriminant validity tests. To evaluate the reliability of the constructs, the values of Cronbach's alpha coefficient (α) and composite reliability (CR) (rho_a and rho_c) were used. Taking into account the criteria of [63, 66], values higher than 0.70 are considered adequate; as shown in Table 3, all the constructs exceeded this threshold. Furthermore, the values of the coefficient of determination (R^2) suggest that AAN, EA, PE, PP, and SI explain 60.1% of the AIO variation.

Discriminant validity was calculated via the criterion of [61], which states that a measurement model presents discriminant validity when the square root of the average variance extracted (AVE) (numbers on the diagonal) should be greater than the correlations with other constructs (numbers outside the diagonal in the same row and column). As shown in Table 3, all the constructs meet this criterion. On the other hand, the heterotrait–monotrait criterion (HTMT) was used, in which all the evaluated constructs have values below the threshold of 0.85 [67], suggesting that the measurement instrument has discriminant validity.

The goodness-of-fit indices of a measurement model are a relevant measure for determining convergent validity [68]; therefore, these criteria provide a reference for researchers to determine to what extent the values obtained fit well with the expected values [62]. Table 4 presents the values of the goodness-of-fit indices of the measurement model, where the standardized root mean

Table 2 Results of the confirmatory factor analysis-CFA

Items		Outer loadings	Standard deviation (STDEV)	*P* values	AVE	Construct	Support
Artificial intelligence improves my academic performance.	PE1	0.781	0.040	0.000	0.680	PE	[45], [64]
Artificial intelligence increases my productivity in my studies.	PE2	0.748	0.040	0.000			
Artificial intelligence is useful for my academic tasks.	PE3	0.872	0.020	0.000			
Artificial intelligence helps me achieve my learning goals faster.	PE4	0.889	0.017	0.000			
People important to me believe that I should use artificial intelligence in my studies.	SI1	0.769	0.033	0.000	0.580	SI	[45], [64]
My teachers support the use of artificial intelligence in learning.	SI2	0.711	0.074	0.000			
My classmates think it is beneficial to use artificial intelligence.	SI3	0.761	0.050	0.000			
In general, the university supports the use of artificial intelligence in education.	SI4	0.813	0.039	0.000			
Using artificial intelligence in my studies is fun.	PP1	0.790	0.047	0.000	0.606	PP	[45], [64]
I enjoy exploring the capabilities of artificial intelligence.	PP2	0.758	0.066	0.000			
I find it exciting to learn with artificial intelligence tools.	PP3	0.819	0.039	0.000			
The use of artificial intelligence makes my studies more entertaining.	PP4	0.743	0.042	0.000			
I am aware of the ethical implications of using artificial intelligence in education.	EA1	0.913	0.015	0.000	0.878	EA	[39]
I find it important to reflect on the ethical aspects of artificial intelligence.	EA2	0.939	0.011	0.000			
I understand the potential ethical dilemmas associated with the use of artificial intelligence in learning.	EA3	0.947	0.008	0.000			
I take ethical considerations into account when using artificial intelligence tools.	EA4	0.947	0.010	0.000			
I feel capable of learning to use artificial intelligence tools.	SL1	0.890	0.021	0.000	0.787	SL	[65]
I am confident in my ability to integrate artificial intelligence into my studies.	SL2	0.922	0.015	0.000			
I can solve problems related to the use of artificial intelligence in my learning.	SL3	0.848	0.026	0.000			
I feel prepared to incorporate artificial intelligence in my studies.	AAN1	0.910	0.015	0.000	0.814	AAN	[65]
I am not concerned about using artificial intelligence tools in my learning.	AAN2	0.915	0.015	0.000			
I am comfortable with the idea of artificial intelligence being part of my education.	AAN4	0.926	0.014	0.000			
I do not experience anxiety at the thought of using artificial intelligence in my academic tasks.	AAN5	0.857	0.030	0.000			
I feel responsible for my use of artificial intelligence in my studies.	AIO1	0.728	0.092	0.000	0.798	AIO	[45], [64]
I feel that I have control over how I use artificial intelligence in my learning.	AIO2	0.822	0.059	0.000			
I make conscious decisions about when and how to apply artificial intelligence in my tasks.	AIO3	0.827	0.046	0.000			
I feel that the integration of artificial intelligence in my studies is up to me.	AIO5	0.755	0.063	0.000			
I appropriate artificial intelligence tools to enhance my educational experience.	AIO6	0.871	0.045	0.000			

square residual (SRMR) presented a value of 0.078, satisfying the threshold proposed by [69]. The value of Chi-square/gl ($\chi 2/df$) shows that the model has an acceptable fit because it has an adequate value between 1 and 3, as proposed by [70]. Finally, the value of the normalized fit index (NFI) is 0.923, which satisfies the threshold of [70].

Testing the research hypotheses

Table 5; Fig. 2 present the main results of the analysis using standardized path coefficients (β), along with p values and confidence intervals for β. Path analysis allows for the determination of β values between the relationships of the exogenous and endogenous variables in the study, as well as the direction of these effects [62, 71]. Hypothesis 3 (H3) revealed a significant effect between PE and AIO, with a path coefficient of $\beta = 0.274^*$ and a p value of $p > 0.006^*$, indicating that PE influences AIO

Table 3 Reliability, discriminant validity, and coefficients of determination

Contruct	α	CR(rho_a)	CR(rho_c)	R^2	AAN	EA	PE	PP	SI	SL	AIO	HTMT
AAN	0.864	0.864	0.917	-	0.902							0.713
EA	0.764	0.784	0.846	-	0.679	0.937						0.723
PE	0.787	0.798	0.86	-	0.621	0.684	0.825					0.755
PP	0.843	0.867	0.894	-	0.560	0.715	0.587	0.778				0.703
SI	0.954	0.957	0.966	-	0.523	0.586	0.681	0.788	0.762			0.654
SL	0.924	0.926	0.946	-	0.811	0.894	0.708	0.689	0.625	0.887		0.739
AIO	0.876	0.921	0.931	0.601	4.250	5.63	0.598	0.621	0.542	0.654	0.752	0.659

Table 4 Model fit

Criteria	Estimated model	Threshold	Author	Decision
SRMR	0.078	< 0.85	[69]	Acceptable
d_ULS	3.212			
d_G	1.322			
χ2/df	2.212	Between 1 and 3	[70]	Acceptable
NFI	0.923	> 0.90	[70]	Acceptable

among university students. Hypothesis 5 (H5) revealed a significant effect between SI and AIO, with a path coefficient of β=0.335* and a p value of $p>0.002$***, demonstrating that SI determines AIO in university students. Hypothesis 6 (H6) shows a significant effect between SL and AIO, with a path coefficient of β=0.431* and a p value of $p>0.029$*, indicating that self-efficacy in learning AI influences AIO in university students. Finally, Hypotheses 1, 2, and 4 did not have significant effects and were thus rejected.

Discussion

The objective of the study was to first test hypothesis H3, which posited that PE significantly influences AIO in university students. This hypothesis was accepted. This finding aligns with previous research that has demonstrated the importance of PE in adopting AI technologies in educational settings. For example, studies by [34] and [39] have shown that PE positively affects students' willingness to accept AI-assisted learning environments and use generative AI products. This is further supported by research highlighting PE as a critical predictor of AI acceptance in higher education [15, 16, 40]. The results of the present study reinforce this trend, suggesting that Peruvian university students are more likely to appropriate AI when they perceive that it will improve their academic performance.

Hypothesis H5, which proposed that SI significantly influences AIO in university students, was also accepted. This result is consistent with previous research highlighting the crucial role of SI in the adoption of AI technologies. For example, the study by [41] on university students in southern India revealed a direct positive effect of SI on the intention to use AI-enabled job application processes. Similarly [42], demonstrated how students adapt their behaviors in response to AI interactions, underscoring the powerful role of social influence. This is further supported by studies showing how social networks and peer influence shape AI adoption patterns [15, 35, 36]. The findings of this study suggest that the social environment of Peruvian university students plays a significant role in their decision to appropriate AI technologies.

Hypothesis H6, which posited that SL significantly influences AIO in university students, was also accepted. This result is in line with those of previous studies demonstrating the importance of self-efficacy in adopting

Table 5 Path, p value, confidence intervals, standard deviation and decision coefficients

Hypothesis		β	*p* value	Percentile		SD	Decision
				2.50%	97.50%		
H_1	**AAN -> AIO**	0.112	0.306	-0.104	0.327	0.109	Rejected
H_2	**EA -> AIO**	-0.185	0.199	-0.472	0.085	0.144	Rejected
H_3	**PE -> AIO**	0.274*	0.006*	0.078	0.471	0.1	Acepted
H_4	**PP -> AIO**	-0.126	0.242	-0.329	0.096	0.108	Rejected
H_5	**SI -> AIO**	0.355*	0.002*	0.135	0.579	0.113	Acepted
H_6	**SL -> AIO**	0.431*	0.029*	0.045	0.816	0.197	Acepted

Note β = path coefficient; SE = standard deviation; ***$p<0.001$; **$p<0.01$; *$p<0.05$

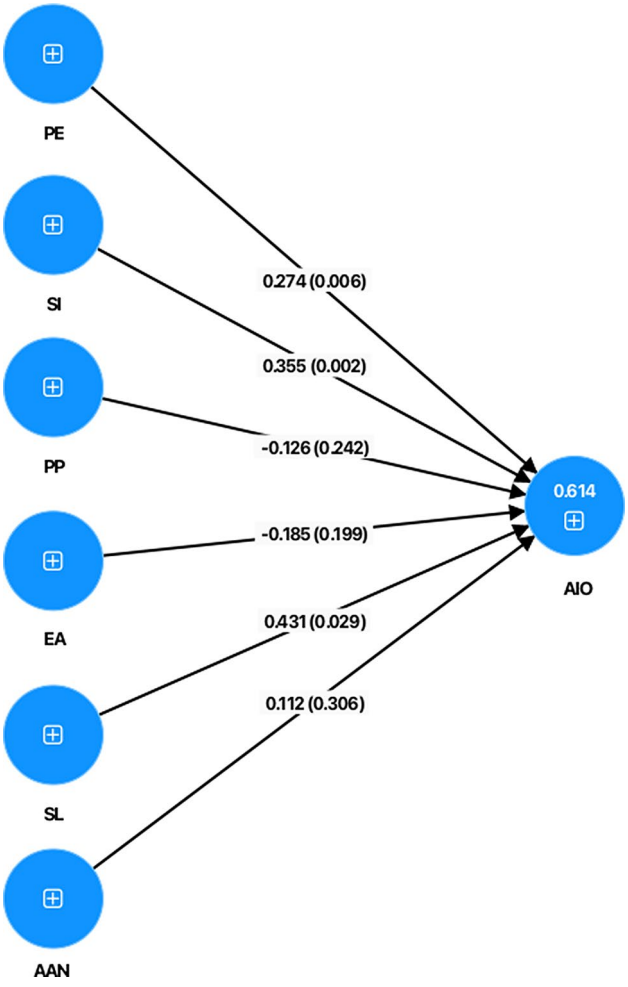

Fig. 2 Resolved research model. Note: At the intersections of the relationship lines are the path coefficients on the left and the p values on the right (inside the parentheses)

AI technologies. For example [37, 52], reported that AI self-efficacy can positively influence students' attitudes toward AI and their actual use of AI tools. The findings of this study suggest that Peruvian university students with greater confidence in their ability to learn and use AI are more likely to appropriate these technologies.

On the other hand, hypotheses H1, H2, and H4 did not have significant effects and were rejected. Hypothesis H1, which proposed that AANs significantly influence AIO, was not supported by the data. This result contrasts with those of previous studies, such as [57], which reported that AI readiness positively influences students' willingness to learn about AI. Other studies have shown how AANs can affect learning motivation and actual AI tool usage [37, 52, 58]. The lack of significance in this study may suggest that, in the Peruvian context, other factors carry more weight in AIO than in readiness and anxiety.

Hypothesis H2, which posited that EAs significantly influence AIO, was also rejected. This result differs from those of studies such as that of [39], who reported that university students' EA significantly influences their behavioral intentions and actual use of generative AI products. Previous research has emphasized the importance of ethical considerations in AI adoption [10, 11, 48, 49]. The absence of a significant effect in this study could indicate that, for Peruvian university students, ethical considerations are not a determining factor in their decision to appropriate AI or that these considerations are overshadowed by more immediate factors.

Hypothesis H4, which proposed that PP significantly influences AIO, was rejected. This finding contrasts with studies such as [45], which emphasize the importance of playful aspects in the adoption of AI technologies. Additional research has demonstrated the role of PP in technology adoption [46, 47]. The lack of a significant effect in this study may suggest that, for Peruvian university students, AIO is more driven by utilitarian factors than by hedonic factors.

In summary, these results offer a nuanced view of the factors influencing AIO among Peruvian university students. While PE, social influence, and self-efficacy in learning AI emerge as key factors, other aspects, such as AI readiness and anxiety, EA, and PP, appear to play a less prominent role in this specific context. These findings underscore the importance of considering contextual and cultural factors when studying the adoption of AI technologies in educational environments.

Theoretical and practical implications

The findings of this study offer significant theoretical and practical implications for AI adoption in higher education. From a theoretical perspective, this research

extends the UTAUT2 model by incorporating psychological constructs specific to AI adoption in educational contexts, contributing to the growing body of literature on technology acceptance models [72, 73]. The study validates the importance of self-efficacy as a crucial factor in technology adoption, aligning with recent research on planned behavior in educational settings [74, 75]. Furthermore, the findings regarding the interplay between performance expectancy and social influence provide new insights into how these factors operate in non-Western educational contexts [76, 79].

The integration of ethical awareness and AI anxiety into the model represents a significant theoretical contribution, particularly in the context of emerging AI technologies such as ChatGPT [76, 77]. This extension of the UTAUT2 model provides a more comprehensive framework for understanding technology adoption in educational settings, especially in developing countries [73, 79]. The study also contributes to the theoretical understanding of how cultural and contextual factors moderate the relationship between psychological factors and technology adoption [78, 79].

From a practical standpoint, the findings provide valuable insights for educational institutions implementing AI technologies. The strong influence of performance expectancy suggests that institutions should focus on demonstrating the tangible benefits of AI tools in improving academic outcomes [75, 76]. The significance of social influence highlights the importance of creating supportive peer networks and fostering a positive institutional culture toward AI adoption [77, 78].

Educational administrators and policymakers can use these findings to develop more effective AI implementation strategies. The study suggests that initiatives should focus on building students' self-efficacy through targeted training programs while addressing anxiety concerns [75, 77]. The findings regarding ethical awareness indicate the need for comprehensive guidelines and educational programs on responsible AI use [76, 78].

Additionally, the study provides practical recommendations for addressing cultural and contextual factors in AI implementation. Institutions in similar cultural contexts can benefit from understanding how these factors influence adoption patterns [79]. The findings suggest that successful AI implementation requires a balanced approach that considers both technological and psychosocial factors [75, 78].

Limitations and future research

The present study has methodological limitations that should be considered when interpreting its findings. The cross-sectional nature of the data collection restricts causal inferences about the relationships between variables [72, 73]. The use of convenience sampling in Peruvian universities may limit the generalizability of findings to other educational contexts [75, 76]. Additionally, the study focused exclusively on student perspectives, without considering the views of educators and administrators, who play crucial roles in AI implementation [77].

These limitations present opportunities for future research directions. Longitudinal studies could provide deeper insights into how AI appropriation evolves over time and how different factors influence adoption patterns at various stages [74, 78]. Cross-cultural comparative studies could help identify universal and context-specific factors in AI adoption across different educational settings [76, 79]. Research incorporating multiple stakeholder perspectives, including those of educators, administrators, and technical staff, would offer a more comprehensive understanding of AI adoption in higher education [77, 78].

Future studies could examine the moderating effects of demographic variables and explore how factors such as academic discipline, previous technology experience, and cultural background influence AI appropriation patterns [75, 79]. An investigation of the relationship between AI appropriation and academic performance outcomes would provide valuable insights for educational policy and practice [76, 78]. Additionally, research exploring the interaction between institutional support systems and individual adoption factors could enhance the understanding of how organizational contexts influence AI appropriation [77, 79].

Conclusions

This research on the factors influencing the appropriation of AI among Peruvian university students reveals three key determinants in the adoption process. PE emerges as a crucial predictor, demonstrating that Peruvian students prioritize tangible benefits in their academic performance when adopting AI technologies. SI proves to be a significant factor, reflecting the collectivist nature of Peruvian culture and its influence on technology adoption decisions. Additionally, SL stands out as a critical determinant, highlighting how students' confidence in their ability to learn and use AI technologies affects their adoption patterns.

The nonsignificant effects of EA, PP, and AAN on AI appropriation provide important insights into the context-specific nature of technology adoption in Peruvian higher education. These findings challenge assumptions about universal adoption factors and emphasize the need for culturally adapted implementation strategies.

The study's methodological approach, which uses an extended UTAUT2 model, offers a refined framework for understanding AI adoption in educational settings. This adaptation, which incorporates context-specific

constructs, contributes to the evolution of technology acceptance models in diverse educational environments.

As AI continues to transform the educational landscape, this research provides evidence-based insights for developing effective AI implementation strategies in higher education, particularly in contexts similar to Peru's educational environment. The findings underscore the importance of considering both technological and cultural factors when introducing AI technologies in educational settings.

Supplementary Information

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Supplementary Material 1

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Data availability
The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
The research that led to these results was approved by the Ethics Committee of the Universidad Nacional de Trujillo and by the University Council by resolution N° 1762–2023/UNT. Informed consent was obtained from all participants included in the study; if participants were under 18 years of age, parents or legal guardians provided informed consent. The intervention was conducted in accordance with the Declaration of Helsinki.

Consent for publication
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